THE PROJECTIVE SCHEMES OVER AN AMPLE BASE FORM A CATEGORY CLOSED UNDER BASE CHANGE, COMPOSITION, FIBRE PRODUCTS, SUMS, AND CONTAIN THE FINITE ONES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 14102-14119 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 14121-14148, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Exact twin in structure of the quasi-projective category card L16110 in this same wave — collected as a separate aggregate per the lane's distinct-statement rule (projective vs quasi-projective categories differ in items and in proof inputs).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The eight-way projective equivalence (lemma-projective, card L16075); Morphisms-chapter base-change-projective, composition-proper, finite-projective; standard ampleness on disjoint unions.

Source reference. Original lemma 0B45: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14028–14044. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02V6: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14148–14148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01W3: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14148–14148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B3I: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14148–14148. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme which has an ample invertible sheaf. Let P_S be the full subcategory of the category of schemes over S satisfying the equivalent conditions of Lemma 0B45.*

(1) *if $S' \to S$ is a morphism of schemes and S' has an ample invertible sheaf, then base change determines a functor $P_S \to P_{S'}$,*
(2) *if $X \in P_S$ and $Y \in P_X$, then $Y \in P_S$,*
(3) *the category P_S is closed under fibre products,*
(4) *the category P_S is closed under finite disjoint unions,*
(5) *if $X \to S$ is finite, then X is in P_S,*
(6) *add more here.*

Proof. Part (1) follows from Morphisms, Lemma 02V6.

Part (2) follows from the fifth characterization of Lemma 0B45 and the fact that compositions of proper morphisms are proper (Morphisms, Lemma 01W3).

If $X \to S$ and $Y \to S$ are projective, then $X \times_S Y \to Y$ is projective by Morphisms, Lemma 02V6. Hence (3) follows from (2).

If $X = Y \amalg Z$ is a disjoint union of schemes and $\mathcal{L}$ is an invertible $\mathcal{O}_X$-module such that $\mathcal{L}|_Y$ and $\mathcal{L}|_Z$ are ample, then $\mathcal{L}$ is ample (details omitted). Thus part (4) follows from the fifth characterization of Lemma 0B45.

Part (5) follows from Morphisms, Lemma 0B3I. □